

Self-Learning: ARM Processor Architecture & Features — MPCA Module 6

Definition: The **ARM (Advanced RISC Machines)** processor is a 32/64-bit **Reduced Instruction Set Computer (RISC)** architecture engineered for high energy efficiency, low power consumption, and high performance in mobile, IoT, and embedded devices.

1. Detailed Technical Explanation

1. Key Architectural Features of ARM:

- Load-Store Architecture:** Data processing instructions (ADD, SUB, AND) operate **only on registers**; memory is accessed exclusively via explicit **LDR** (Load) and **STR** (Store) instructions.
- Inline Barrel Shifter:** One operand in any arithmetic/logic instruction can be shifted/rotated by arbitrary bits in **hardware in the same clock cycle** without extra instructions.
- Example: **ADD R0, R1, R2, LSL #2** ($R0 = R1 + R2 \times 4$).
- Conditional Execution of All Instructions:** Every ARM instruction contains a 4-bit condition code field (**EQ** , **NE** , **GT** , **LT** , **AL**), eliminating unnecessary jump/branch instructions in small **if-else** blocks.
- Dual Instruction Sets (ARM vs Thumb):**
 - **ARM State (32-bit):** Full 32-bit high-performance instruction set.
 - **Thumb State (16-bit):** Compact 16-bit instruction set offering up to **30-40% code density reduction** for memory-constrained microcontrollers.

2. Register Organization of ARM (ARM7 / Cortex-A)

- Total of **37 registers** (31 General-Purpose 32-bit registers and 6 Status registers), organized into banked registers across operating modes:
- **R0 - R12** : General-purpose data registers.
- **R13 (SP)** : Stack Pointer.
- **R14 (LR)** : Link Register (stores return address upon function call).
- **R15 (PC)** : Program Counter.
- **CPSR** : Current Program Status Register.
- **SPSR** : Saved Program Status Register (used during interrupt/exception modes).

3. CISC (x86/Pentium) vs RISC (ARM) Comparison

Feature	Intel x86 / Pentium (CISC)	ARM Processor (RISC)
Instruction Size	Variable (1 to 15 bytes)	Fixed (32-bit ARM / 16-bit Thumb)
Memory Access	Allowed directly in arithmetic ops	Strict Load-Store Only (LDR/STR)
Power Consumption	Higher (Desktop / Server oriented)	Ultra Low Power (Battery / Mobile)
Pipelining	Complex hyper-pipelines (20-31 stages)	Simple, highly efficient (3-8 stages)
Industry Dominance	PC, Laptops, Cloud Servers	Smartphones (99% iOS/Android), IoT, Apple Silicon (M1/M2/M3)

4. Quick Recall Flow

ARM RISC -> Load-Store Architecture -> Inline Barrel Shifter -> Conditional Execution -> ARM (32-bit) / Thumb (16-bit) -> Mobile & Embedded Dominance